Kazakh-British Technical University

Data collection and preparation

STUDENT'S INDIVIDUAL WORK (SIS) №1

Data Collection, Preparation, and Analysis: R&B Artists

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INTRODUCTION

This report covers topics about collecting, processing and analyzing information about alternative R&B music artists using Python. Data was collected from two different sources: the MusicBrainz REST API and Wikipedia's list of alternative R&B artists. Python libraries such as Requests, BeautifulSoup, and Pandas were used for data collection, cleaning, merging, and exploratory data analysis (EDA). Matplotlib was used to create visualizations showing the geographical distribution, artist types, and birth or formation decades of alternative R&B performers. The project aims to demonstrate how information from different sources can be combined to identify patterns and explore the characteristics of artists within the alternative R&B genre.

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Task 1. Data Collection

Here we retrieve and collect data:

```
Topic: Music R&B
API:https://musicbrainz.org/ws/2/artist/
ws:https://en.wikipedia.org/wiki/List_of_alternative_R%26B_artists?
```

```
#step1
#collecting data

import requests
import pandas as pd
from bs4 import BeautifulSoup
import time
wiki_url="https://en.wikipedia.org/wiki/List_of_alternative_R%26B_artists"
api_url="https://musicbrainz.org/ws/2/artist/"

wiki_headers={"User-Agent": "StudentMusicResearch/1.0 (educational project)"}
response=requests.get(
    wiki_url,
    headers=wiki_headers,
    timeout=20)
print("Status:", response.status_code)
response.raise_for_status()
soup=BeautifulSoup(response.text, "html.parser")
print("Page title:", soup.title.get_text(strip=True))
```

```
Status: 200
Page title: List of alternative R&B artists - Wikipedia
```

```
columns=[
    "Wiki_Artist",
    "Artist",
    "MusicBrainz_ID",
    "Country",
    "Artist_Type",
    "Start_Date",
    "Ended",
    "Match_Score"]
df_api=pd.DataFrame(music_records, columns=columns)
print("MusicBrainz dataset size:", df_api.shape)
df_api.head(10)
```

```
MusicBrainz dataset size: (14, 8)
```

	Wiki_Artist	Artist	MusicBrainz_ID	Country	Artist_Type	Start_Date	Ended	Match_Score
0	6lack	6LACK	07832b42-8826-4ab1-acd3-c49a2f595ffe	US	Person	1992-06-24	None	100
1	Aaliyah	Aaliyah	bc85da58-52d9-457d-ae8d-5d8d4ec870a9	US	Person	1979-01-16	True	100
2	Abra	ABRA	b7c20887-5042-451d-ac3c-772e25cbe9d8	US	Person	1989-03-30	None	100
3	Jhené Aiko	Jhené Aiko	ceb3f5b7-9f1a-430f-afbe-976fdcf30482	US	Person	1988-03-16	None	100
4	AlunaGeorge	AlunaGeorge	e9da3ed5-eb9a-4f70-a0ed-4e59e380601f	GB	Group	2009	None	100
5	Autre Ne Veut	Autre Ne Veut	ef971f23-57a5-4e4d-a757-37dbf3937ab5	US	Person	1982-04-20	None	100
6	Erykah Badu	Erykah Badu	321531fc-db73-4ffa-a959-61a61a2908c1	US	Person	1971-02-26	None	100
7	Banks	BANKS	faa28a33-5470-4b90-90c3-e71955d93a44	US	Person	1988-06-16	None	96
8	Blood Orange	Blood Orange	3d254f1e-7d18-431b-baeb-68e57695dfdb	GB	Person	1985-12-23	None	100
9	Jesse Boykins III	Jesse Boykins III	2fbc5d12-d30b-41ab-a5ce-042aa60fed38	US	Person	1985-02-20	None	100

In this project, I selected Alternative R&B Music Artists Analysis as my research topic. I collected artist names and Wikipedia links from the Wikipedia article "List of alternative R&B artists" using Requests and BeautifulSoup. I also used the MusicBrainz REST API to retrieve additional artist information, including country, artist type, activity dates, and MusicBrainz identifiers. Both datasets were organized into Pandas DataFrames and saved as CSV files for further cleaning, merging, and analysis.

Task 2. Data Cleaning

We work with cleaning and preparing data for further analysis:

```
df_wiki=df_wiki.dropna(subset=["Artist_Name"]).copy()
df_api=df_api.dropna(subset=["Artist_Name"]).copy()

df_wiki=df_wiki[df_wiki["Artist_Name"] != ""].copy()
df_api=df_api[df_api["Artist_Name"] != ""].copy()

df_api["Country"]=df_api["Country"].fillna("Unknown")
df_api["Type"]=df_api["Type"].fillna("Unknown")

df_api["Begin_Date"]=df_api["Begin_Date"].fillna("")
print("Missing values after cleaning:")
print(df_api.isnull().sum())
```

```
api missing values:
Wiki_Artist          0
Artist               0
MusicBrainz_ID       0
Country              1
Artist_Type          1
Start_Date           1
Ended               13
Match_Score          0
dtype: int64
```

```
Missing values after cleaning:
Wikipedia_Artist     0
Artist_Name          0
MBID                 0
Country              0
Type                 0
Begin_Date           0
Ended               13
Search_Score         0
Begin_Year           1
Artist_Key           0
dtype: int64
```

So I cleaned the Wikipedia and MusicBrainz datasets using the Pandas library. I checked the datasets for missing values and duplicate records, standardized artist names, and renamed columns to improve consistency. Missing country and artist type information was labeled as "Unknown", while unavailable dates were preserved without inventing values. Finally, I corrected data types, extracted beginning years where available, and saved the cleaned datasets for merging and analysis.

Task 3. Datasets Merging

Merging the datasets using merge function and matching columns:

```
df_merged=pd.merge(
    df_wiki,
    df_api.drop(columns=["Artist_Name"]),
    on="Artist_Key",
    how="left",
    indicator=True,
    validate="one_to_one")
print("Merged df:")
display(df_merged.head(10))
print("\nMerged df shape:", df_merged.shape)
```

Merged df:

	Artist_Name	Wikipedia_Link	Artist_Key	Wikipedia_Artist	MBID	Country	Type	Begin_Date	Ended	Search_Score	Begin_Year
0	070 Shake	https://en.wikipedia.org/wiki/070_Shake	070 shake	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	6lack	https://en.wikipedia.org/wiki/6lack	6lack	6lack	07832b42-8826-4ab1-acd3-c49a2f595ffe	US	Person	1992-06-24	NaN	100.0	1992.0
2	Aaliyah	https://en.wikipedia.org/wiki/Aaliyah	aaliyah	Aaliyah	bc85da58-52d9-457d-ae8d-5d8d4ec870a9	US	Person	1979-01-16	True	100.0	1979.0
3	Abra	https://en.wikipedia.org/wiki/Abra_(singer)	abra	Abra	b7c20887-5042-451d-ac3c-772e25cbe9d8	US	Person	1989-03-30	NaN	100.0	1989.0
4	Jhené Aiko	https://en.wikipedia.org/wiki/Jhené_Aiko	jhené aiko	Jhené Aiko	ceb3f5b7-9f1a-430f-afbe-976fdcf30482	US	Person	1988-03-16	NaN	100.0	1988.0
					e9da3ed5-						

We merged the cleaned Wikipedia and MusicBrainz datasets using Pandas. I standardized the artist names and created matching keys to connect records from both sources. A left join was used to preserve all Wikipedia artists while adding available MusicBrainz information such as country, artist type, and beginning year. Finally, I examined matched and unmatched records, checked the merged dataset for duplicates and missing values, and saved the combined dataset as rnb_merged.csv.

Task 4. Data Analysis

Here is the analysis:

Desc. stats:

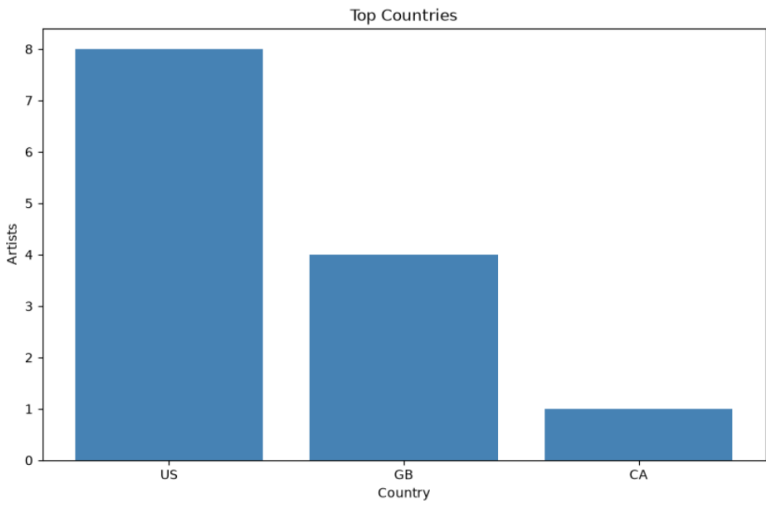
	Begin_Year	Search_Score
count	13.000000	14.000000
mean	1987.615385	97.857143
std	8.893241	6.137616
min	1971.000000	77.000000
25%	1985.000000	100.000000
50%	1988.000000	100.000000
75%	1989.000000	100.000000
max	2009.000000	100.000000

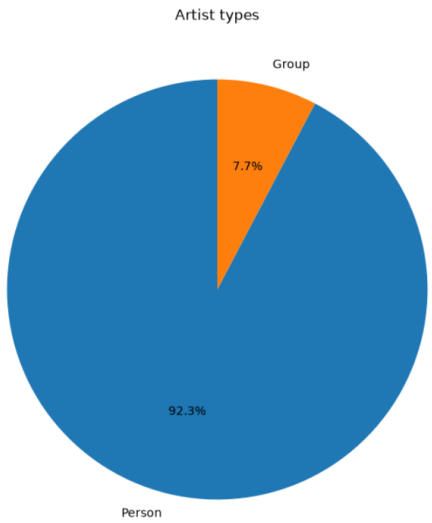
	Metric	Value
0	Total Wikipedia artists	88
1	Matched MusicBrainz artists	14
2	Unmatched artists	74
3	Countries represented	4
4	Individual artists	12
5	Musical groups	1

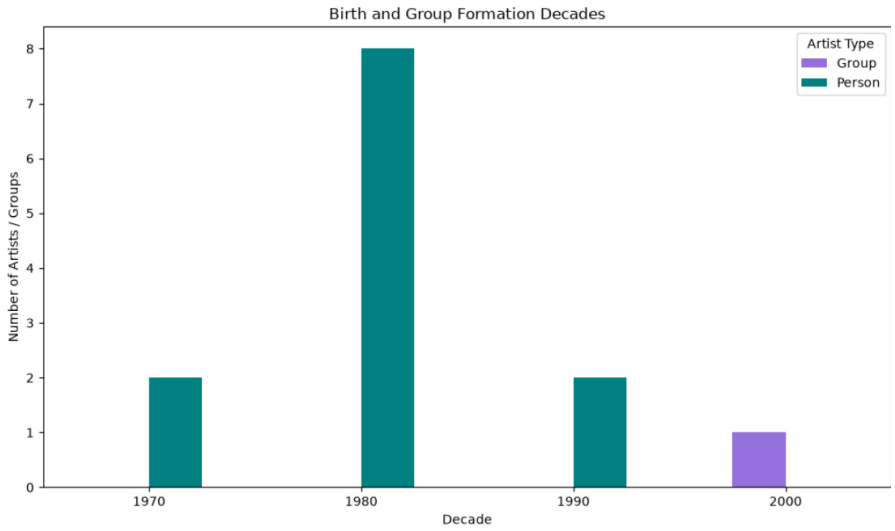
Now we are performing exploratory data analysis on the merged alternative R&B artist dataset using Pandas. I examined the distribution of artists by country and artist type, calculated descriptive statistics, and investigated available birth and group formation years. I also calculated a Pearson correlation between birth years and MusicBrainz search relevance scores for individual artists, treating the result as exploratory rather than an indicator of popularity. Finally, I summarized the main characteristics of the dataset and prepared the results for visualization.

Task 5. Visualisation

So we are practicing with visualizing the data:







We are using Matplotlib to visualize the alternative R&B artist dataset collected from Wikipedia and MusicBrainz. I created charts to examine the geographical distribution of artists, the proportions of different artist types, and the distribution of artist birth and group formation decades. Each visualization included a descriptive title and appropriate labels to make the results easier to interpret. The charts were saved as PNG images for inclusion in the final project report.

CONCLUSION

This project demonstrated the complete process of collecting, preparing, analyzing, and visualizing real-world music data using Python. Information from Wikipedia and the MusicBrainz API was cleaned and merged into a structured dataset, allowing the geographical distribution, artist types, and available biographical information to be examined. Descriptive statistics, correlation analysis, and visualizations were used to explore the collected data and communicate the results. The project also highlighted challenges such as missing values, incomplete API responses, and differences between data sources. Overall, this work improved practical skills in web scraping, API integration, data cleaning, exploratory data analysis, and data visualization.